

Cache walk

A direct-mapped cache splits each address into tag, index, and offset

Address 0x14 starter

- Starter: cold direct-mapped cache
- Address hex fourteen decodes to tag one, index one, offset zero
- First Access is a miss; install loads the line into set one
- Access again is a hit
- Address hex fifteen shares tag and index with offset one, still a hit, spatial locality
- Address hex twenty-four shares index one but tag two, a conflict miss

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Cache hit/miss walkthrough

Split an address into tag | index | offset, probe a tiny direct-mapped cache, and see hit vs miss (then install on miss). Starter: 8-bit addresses, 4 sets × 4-byte lines — access 0x14.

Starter example: cold direct-mapped cache, addr 0x14 (tag=1, index=1, offset=0). Access → miss + install; Access again → hit.

Challenge 0 / 22 cleared

Quiz: three fields: A cache address is commonly split into...

☐ tag, index (set), and offset (within line)

☐ only a byte enable

☐ opcode and funct

☐ row and column of a DRAM only

Show hint

Check

Next

Load challenge setup

Idle

Quiz: three fields

Quiz: index

Quiz: tag

Quiz: offset

Quiz: direct-mapped

Quiz: hit

Quiz: conflict

Quiz: line size

Decode 0x14

First access miss

Repeat hit

Same line 0x15

Conflict miss

Decode 0x00

Cache walk starter

learn_digital — cache walk

Workbook practice

- On paper, split address hex fourteen with two index and two offset bits
- Label tag, index, and offset values
- Draw four sets with one line each
- Trace cold access, install, repeat hit, and one conflict miss
- Compute line size as two to the offset bits and capacity as sets times line size
- Name one pitfall: assuming same index means same tag

Pitfalls to watch

- Do not treat index as the full address, tag distinguishes lines in the same set
- Valid zero always means miss
- Direct-mapped is simple but thrashes on conflicts
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, decode zero fourteen, miss once, hit twice, then try zero twenty-four
- On paper, fill one row of the set table after install
- When you are ready, take the short quiz, then continue to dual-port RAM